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Assessing urban water security in Awash River Basin, Addis Ababa, and its surrounding towns using Integrated Urban Water Security Index

Abebe M. Legass^{1*} , Tena Alamirew^{1,2}, Solomon G. Gebrehiwot^{1,3}, David Haro-Monteagudo⁴, Lewoye Tsegaye⁵ and Getnet Bogale Begashaw⁶

*Correspondence:

Abebe M. Legass

abebemengistu@gmail.com

¹School of Civil and Environmental Engineering, Addis Ababa University, Addis Ababa, Ethiopia

²Water and Land Resource Center, Addis Ababa, Ethiopia

³Inst. Landscape Ecology and Resource Management, Justus-Liebig University, Giessen, Germany

⁴Geography and Environment, School of Geosciences, University of Aberdeen, Aberdeen AB24 3UF, UK

⁵College of Agriculture and Environmental Science, Department of Natural Resource Management, Bahir Dar University, Bahir Dar, Ethiopia

⁶Department of Data Science, College of Natural and Computational Science, Debre Berhan University, P.O. Box 445, Debre Berhan, Ethiopia

Abstract

This study assesses the urban water security of Addis Ababa and surrounding cities, within the Awash River Basin (ARB). Using the Integrated Urban Water Security Index (IUWSI) and Analytic Hierarchy Process (AHP), it combined primary data from household surveys across Addis Ababa's 11 sub-cities with stakeholder perspectives from downstream towns (Adama, Awash, Lake Beseka) and secondary data from relevant authorities and literature. The analysis applied the framework of Drinking Water and Human well-being, Ecosystems, Climate Change and Water-Related Hazards, Socio-Economic Aspects (DECS). Statistical analyses (chi-square, regression, K-means clustering) confirmed significant correlations between location, water availability, and affordability. ArcGIS spatial mapping revealed distributions of turbidity (30 NTU downstream), *E. coli* (50 CFU/100 mL), and Water Quality Index (40–80). Water shortages affected 79.2% of respondents in Addis Ababa and 90–100% downstream, with Lake Beseka recording arsenic at 95.6 µg/L. Bole (IUWSI: 2.1) and Akaki Kaliti (2.2) were the most vulnerable sub-cities. Future projections indicate IUWSI could fall to 1.6 under Business-as-Usual or rise to 3.9 by 2050 under Sustainable Development scenarios. Recommendations include infrastructure upgrades, pollution control, climate-resilient strategies, transboundary governance, and enhanced community engagement, providing a scalable model for urbanizing basins and supporting Sustainable Development Goal 6 (SDG 6).

Keywords Water Security, Awash River Basin, IUWSI, DECS Framework, Water Quality, Upstream-Downstream Dynamics, SDG 6, Addis Ababa

1 Introduction

Water security is defined as the sustainable availability of adequate quantities and quality of water for human, ecological, and economic needs with acceptable risks [1]. Water security is critical for urban resilience and achieving SDG 6 for clean water and sanitation [2].



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Water security frameworks, such as DECS [3], City Blueprint [4], and others [5, 6], emphasize multidimensional assessments encompassing water supply, sanitation, ecosystems, and socio-economic factors.

In Addis Ababa, [7] developed a household-scale framework focusing on water supply and sanitation, but overlooked ecosystem degradation and climate impacts. Climate variability, with increased rainfall variability and two droughts per decade, reduces water availability [8, 9]. Socio-economic disparities, particularly affordability, disproportionately burden low-income households [10].

Globally, urban water security studies in Nairobi [11] and Lagos [12] highlight similar challenges, but few integrate upstream–downstream dynamics or water quality parameters like turbidity and *E. coli*. Regionally, the Nile Basin's flood management and Abbay Basin's groundwater quality [13] offer comparative insights. Locally, the Rift Valley Lakes' lower industrial impacts [14] contrast with the ARB's challenges. This study advances literature by applying the DECS framework, integrating primary and secondary data, and employing advanced statistical and spatial analyses to address water security challenges.

Addis Ababa, Ethiopia's capital within the Awash River Basin (ARB), faces escalating water security challenges due to a population exceeding 5.4 million (projected to reach 9 million by 2030, 4% annual growth) [15], rapid industrialization, climate variability (1.5 °C warming by 2030), and aging infrastructure [16]. The ARB, spanning 106,522.8951 km², supports over 18.6 million people, including downstream communities, but is stressed by upstream pollution (30% wastewater treatment in Addis Ababa), downstream heavy metal contamination (arsenic: 95.6 µg/L at Lake Beseka), and a 40% flow reduction due to over-extraction [9, 17].

Addis Ababa, contributing 29% of Ethiopia's GDP, relies on surface water (Legedadi Reservoir, Gefersa and Dire) and groundwater, managed by Addis Ababa Water and Sewerage Authority (AAWSA), but meets only 40% of its 1.2 million m³/day demand due to 45% non-revenue water losses [18]. Downstream, pastoralist communities and irrigation schemes face acute shortages and health risks from industrial and geothermal pollution. Upstream activities, such as untreated sewage and industrial effluents, degrade downstream water quality, creating allocation tensions [17]. Climate variability exacerbates these issues, mobilizing contaminants and reducing flows, with 30% of the basin at flood risk and 70% of downstream irrigation schemes water-stressed [9].

This study employs the DECS framework [3] to assess water security across four dimensions: Drinking water and human well-being, ecosystems, climate change and water related hazards, and socio-economic aspects. By integrating primary data from a 2025 questionnaire survey (n=385) in Addis Ababa's 11 sub-cities and secondary sources, the study quantified Addis Ababa's water security. The AHP, a multi-criteria decision-making tool, prioritizes indicators based on stakeholder inputs [19], while ArcGIS enables spatial visualization of disparities using shapefile data. The IUWSI and AHP provide a robust methodology, while ArcGIS maps spatial disparities. Statistical analyses (chi-square, regression, K-means clustering) identified key drivers. While previous studies in Addis Ababa have focused on household-scale water supply [7]) or specific water quality parameters [17], a critical gap remains in the integrated, spatial assessment of urban water security that connects upstream urban centers with downstream rural and ecological systems within a basin context. This study addresses this gap by introducing

a novel application of the Integrated Urban Water Security Index (IUWSI) within the DECS framework to the Awash River Basin. The study approach uniquely combines high-resolution spatial mapping, advanced statistical clustering (K-means), and future scenario projections to not only quantify current disparities but also to model the long-term efficacy of potential interventions. This provides a scalable, evidence-based decision-support tool for policymakers tasked with achieving SDG 6 in rapidly urbanizing river basins.

Hence, this study aims to quantify Addis Ababa's water security status using the IUWSI and AHP, identify key water-related challenges impacting economic and human development, spatially map water security and quality parameters (IUWSI, turbidity, *E. coli*, WQI), and project future scenarios (2030-2050) while proposing evidence-based strategies.

This research advances prior studies [7, 20] by integrating macro-level drivers (climate change, ecosystems) and water quality parameters (turbidity, *E. coli*, WQI), thereby establishing a comprehensive model applicable to urbanizing basins worldwide.

2 Methodology

This study employed a mixed-methods approach to assess urban water security in the Awash River Basin, focusing on Addis Ababa and downstream areas, using the DECS framework (Drinking Water and Human Well-being, Ecosystems, Climate Change and Water-Related Hazards, Socio-Economic Aspects) and the Integrated Urban Water Security Index (IUWSI). Primary data were collected via a 2025 questionnaire survey ($n=385$) across Addis Ababa's 11 sub-cities, targeting households, community leaders, and officials, supplemented by stakeholder insights from downstream towns (Adama, Awash, Lake Beseka). Secondary data on water supply, quality, hydrology, and socio-economics were sourced from agencies (AAWSA, Basin Development Authority, CSA) and literature, with water quality parameters (e.g., arsenic, turbidity, *E. coli*) standardized against national and WHO guidelines. Climate projections involved bias correction (Quantile Mapping) and ensembling of three CMIP6 GCMs (ACCESS, MRI-ESM2-0, NorESM2-MM) for temperature and precipitation under SSP2-4.5 and SSP5-8.5 scenarios. Indicators were normalized (min–max to 1–5 scale), weighted via Analytic Hierarchy Process (AHP) with 20 stakeholders (consistency ratio <0.1), and aggregated into IUWSI scores. Analyses included descriptive statistics, chi-square tests, OLS regression, K-means clustering (three clusters), and ArcGIS spatial mapping (using shapefiles for choropleth and kernel density visualizations of IUWSI, turbidity, *E. coli*, and WQI). Future projections (2030-2050) compared Business-as-Usual (BAU) and Sustainable Development (SD) scenarios to evaluate potential IUWSI trajectories.

2.1 Study Area

The Awash River Basin (ARB) is Ethiopia's second-largest basin (106,522.89 km²), spanning from the highlands ($>3,000$ m a.s.l.) to the Afar lowlands. The ARB can be defined with or without the Eastern sub-catchment. Including the Eastern sub-catchment, the basin terminates at Lake Abhe (on the Ethiopia-Djibouti border). Excluding it, the main river ends at Lakes Afambo, Abi, and Gemeri, respectively. This study includes the Eastern sub-catchment. The main river originates in the Chilimo highlands near Ginchi, while the Eastern sub-catchment starts in the Hararghe Highlands. The basin's

topography is 48.1% flat and 51.9% variable steep terrain, with elevations ranging from 250 m a.s.l. near Afambo to 4,178 m a.s.l. at Chilalo Mountain in the Arsi Zone's Hitosa district (Fig. 1).

It includes upstream Addis Ababa (8.8°–9.1°N, 38.7°–38.8°E), cities of Debre Zeit (8.75°N, 38.98°E) and Adama (8.54°N, 39.27°E), midstream cities including Metehara (8.90°N, 39.917°E), Awash Town (8.991°N, 40.165°E), Lake Beseka (8.86°N, 39.87°E), downstream cities Melka Werer (9.5°N, 40.5°E), Dubti (11.72°N, 41.07°E), and the terminal delta near Aysaita (11.57°N, 41.44°E). Addis Ababa consists of 11 sub-cities: Addis Ketema, Akaki Kality, Arada, Bole, Gullele, Kirkos, Kolfe Keranio, Lemi Kura, Lideta, Nifas Silk-Lafto, and Yeka. The basin features bimodal rainfall (1600 mm upstream, 160 mm downstream) and 70% agricultural land use, contributing to sedimentation and pollution [17].

2.2 Data and Data Collection

A questionnaire survey (May 15–June 22, 2025) targeted 385 respondents in Addis Ababa: 331 households (86.1%), 38 community leaders (9.9%), and 16 AAWSA officials (4.0%), selected via stratified random sampling across 11 sub-cities (Appendix A: Questionnaire Summary). Questions covered water sources, availability (hours/day), quality perception (Poor to Excellent), affordability, contamination awareness, and governance satisfaction.

The study employed a comprehensive analysis of secondary data to establish the hydrological, infrastructural, and environmental context of the basin. Core water supply and sanitation metrics indicating 78% water supply coverage, 65% sanitation access, and a notably high 45% non-revenue water were procured from the Addis Ababa Water and Sewerage Authority [21], highlighting significant system inefficiencies. The broader

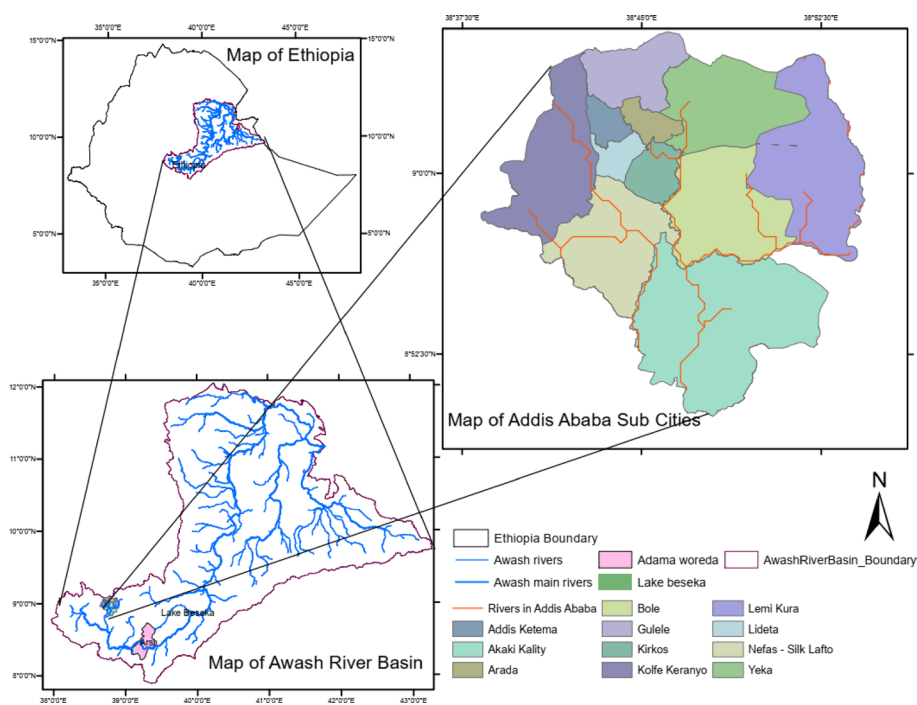


Fig. 1: Location of Study Area. Sub-cities of the Addis Ababa City in the right panel, the Awash River Basin in the left panel

hydrological framework, including a basin-wide flow of 4.6 billion m³/year and key population statistics [22], provided the essential scale for the analysis.

Water quality assessment was guided by national standards [23], which define critical thresholds for parameters such as turbidity (5–30 NTU) and *E. coli* contamination (0–50 CFU/100 mL). To populate these parameters with site-specific values, the study synthesized recent empirical findings from the literature. This included integrated water quality data, encompassing biochemical oxygen demand (BOD: 15–20 mg/L) and heavy metal concentrations, from 20 sampling stations (SW1–SW20) strategically located along the river's course from upstream (Addis Ababa/Koka) to downstream (Lake Beseka) [17]. Further context on organic contaminants [24]) and long-term climate impacts [9]) was also incorporated to ensure a robust multi-pollutant and multi-stressor perspective.

2.3 Climate Data Processing

2.3.1 Bias Correction and Ensembling Methods

To address systematic biases in General Circulation Model (GCM) outputs and combine multiple model predictions, the study applied Quantile Mapping (QM) for bias correction and a simple averaging ensemble method to daily maximum (Tmax) and minimum (Tmin) temperature data for Addis Ababa.

GCMs ACCESS, MRI-ESM2-0, and NorESM2-MM were chosen based on their proven performance in capturing the Awash River Basin's (ARB) unique climatic characteristics, as demonstrated in prior regional climate studies [25, 26]. The ACCESS was selected for its high resolution and ability to simulate tropical rainfall patterns, critical for the ARB's monsoon-driven bimodal rainfall (1,600 mm upstream, 160 mm downstream). The MRI-ESM2-0 excels in representing East African climate dynamics, particularly the variability of the Inter-Tropical Convergence Zone (ITCZ), which governs the ARB's rainfall distribution [7, 25, 26]. NorESM2-MM was chosen for its robust simulation of temperature extremes and evapotranspiration, essential for modeling water availability in the ARB's semi-arid downstream regions [25, 26, 27]. These models were evaluated for their ability to reproduce historical climate data in the ARB, ensuring reliable bias-corrected projections of daily maximum and minimum temperatures (Tmax, Tmin) under CMIP6 scenarios, which are critical for assessing climate change impacts on water security in the context of a projected 1.5 °C warming by 2030 [16].

These methods ensure that GCM outputs align with observed climate data and provide a robust, combined estimate of future climate conditions.

2.3.2 Bias Correction: Quantile Mapping

Quantile Mapping (QM) is a statistical technique used to correct biases in GCM outputs by aligning the distribution of modeled data with that of observed data [28, 29]. The method adjusts the GCM data to match the statistical properties of the observed data while preserving the temporal structure of the model output. The QM process is implemented as follows:

- i. **Data Alignment:** For each GCM (ACCESS, MRI-ESM2-0, NorESM2-MM), daily Tmax or Tmin data were aligned with observed data based on a common time index ('time' column in Excel files) using a merge operation to ensure temporal correspondence.

ii. Cumulative Distribution Function (CDF) estimation: For each paired dataset (observed and GCM predicted), temperature values ('tasmax' for Tmax, 'tasmin' for Tmin) were used to estimate empirical CDFs. A normal distribution was assumed, with parameters (mean μ and standard deviation σ) derived from the datasets. The CDF for observed data, $F_{obs}(x)$, and GCM data, $F_{gcm}(x)$, were calculated as:

$$F_{(x)} = \Phi\left(\frac{x - \mu}{\sigma}\right), \quad (1)$$

where Φ is the standard normal CDF.

iii. Quantile Mapping: GCM temperature values were transformed to match the observed distribution. For a GCM temperature x_{gcm} , its quantile was computed as $u = F_{gcm}(x_{gcm})$. This quantile was mapped to the observed distribution using the inverse CDF:

$$x_{corrected} = F_{obs}^{-1}(u) = \mu_{obs} + \sigma_{obs} \cdot \Phi^{-1}(u) \quad (2)$$

This ensures the corrected GCM data have the same statistical distribution as the observed data. iv. Output: The bias-corrected GCM data were stored as time series with the original 'time' column and corrected temperature values ('tasmax' or 'tasmin').

QM was applied separately to Tmax and Tmin datasets for each GCM, resulting in bias-corrected outputs suitable for climate impact studies [27].

2.3.3 Ensembling: Averaging

To combine bias-corrected outputs from multiple GCMs, averaging ensemble method was employed to reduce model-specific uncertainties [30]. The process was as follows:

1. Data Alignment: Bias-corrected Tmax and Tmin datasets from all GCMs were aligned by their 'time' column using an inner merge to include only common time periods.

2. Mean Calculation: For each day, the ensemble mean was calculated as the arithmetic average of the bias-corrected temperature values across the GCMs:

$$T_{ensemble}(t) = \frac{1}{N} \sum_{i=1}^N T_{corrected,i}(t) \quad (3)$$

where N is the number of GCMs (three: ACCESS, MRI-ESM2-0, NorESM2-MM), and $T_{corrected,i}(t)$ is the bias-corrected temperature for model i at time t .

3. Output: The ensemble mean was saved as a time series with columns 'time' and 'Ensemble Mean' for Tmax and Tmin.

The simple averaging approach assumes equal weighting for each GCM, suitable when model performance is comparable [31].

Implementation: The methods were implemented in Python using the pandas, numpy, and scipy libraries. Input Excel files contained 'time' and temperature columns ('tasmax' or 'tasmin'). Outputs included ensemble means and individual bias-corrected datasets saved as Excel files.

2.4 Selection of Water Quality Parameters

Six water quality parameters (Arsenic ($\mu\text{g/L}$), Manganese ($\mu\text{g/L}$), Emerging Organic Contaminants (EOCs, $\mu\text{g/L}$), Biochemical Oxygen Demand (BOD, mg/L), Turbidity (NTU), and *E. coli*) were selected for their alignment with the DECS framework's dimensions (Drinking Water, Ecosystems, Socio-Economic) and their relevance to the Awash River Basin's (ARB) urban and downstream challenges. Arsenic and *E. coli* were selected as they represent critical, yet distinct, health risks; arsenic for chronic toxicity (e.g., skin lesions, cancers) and *E. coli* for acute waterborne diseases (e.g., diarrhea) [32]. While Manganese can occur naturally, elevated levels in urban settings like Bole sub-city ($100 \mu\text{g/L}$) are strongly indicative of industrial and municipal wastewater pollution [33]. Similarly, BOD, though originating from various organic sources, serves as a key aggregate indicator of pollution load from untreated sewage and industrial effluents, which are predominant in the ARB [34], while EOCs indicate emerging urban threats [35, 36]. Turbidity signals pathogen risks and sedimentation from agricultural runoff, impacting water treatment [37]. These parameters, prioritized via AHP, enable spatial mapping of pollution hotspots and inform SDG 6 interventions, addressing the ARB's rapid urbanization and limited wastewater treatment.

2.4.1 Data Integration and Standardization

To ensure reliability and consistency across diverse secondary data sources, a rigorous standardization protocol was applied. All water quality data (e.g., from AAWSA, CSA, and literature) were cross-referenced against Ethiopian national standards to ensure comparable measurement units and methodologies. Socio-economic data from the questionnaire were designed to align with Central Statistics Agency (CSA) census categories. Temporal discrepancies were mitigated by focusing on the most recent data and using them to represent baseline conditions for the 2023–2025 study period. For older but critical datasets (e.g., [9] on flow reduction), the findings were used to establish historical trends rather than absolute current values. They were supplemented with recent agency reports (e.g., BDA, 2022).

2.4.1.1 Data Analysis The DECS framework guided indicator selection, normalized using the min–max method to a 1–5 scale:

$$S_i = \left(\frac{V_i - V_{\min}}{V_{\max} - V_{\min}} \right) (S_{\max} - S_{\min}) + S_{\min} \quad (4)$$

where V_i represents the observed value of the variable i for which $V_{\min}$ and $V_{\max}$ are the minimum and maximum possible values respectively. $S_{\min}$ and $S_{\max}$ correspond to the scale maximum and minimum values i.e., 1 and 5.

$$S_i = \left(1 - \frac{V_i - V_{\min}}{V_{\max} - V_{\min}} \right) (S_{\max} - S_{\min}) + S_{\min} \quad (5)$$

Subsequently, each indicator, I_j is estimated by calculating the weighted arithmetic mean (next Eq. 3) of the standardized score of all the corresponding variables. Likewise, each dimension, D_k will be computed by taking a weighted arithmetic mean (Eq. 7) of all the indicators in that particular dimension

$$I_j = \frac{\sum_{i=1}^n S_i * w_i}{\sum_{i=1}^n w_i} \quad (6)$$

$$D_k = \frac{\sum_{j=1}^m I_j * w_j}{\sum_{j=1}^m w_j} \quad (7)$$

where w_i and w_j are the weights assigned to variables and indicators respectively.

Scores corresponding to all five dimensions will be averaged to calculate the UWSI (Urban Water Security Index) for the main cities in the Awash River Basin (Eq. 8).

$$UWSI = \frac{1}{4} \sum_{k=1}^4 D_k \quad (8)$$

UWSI scores for Addis Ababa city in the Awash River Basin be interpreted based on the above details.

In the AHP methodology, 20 stakeholders were selected to represent diverse perspectives across the Awash River Basin, including water management experts, community leaders, and local officials from upstream (Addis Ababa) and downstream areas (Adama, Awash Town, Lake Beseka). Stakeholders were chosen via purposive sampling based on their expertise in water governance or community representation, ensuring balanced input [19]. Pairwise comparisons were conducted through structured interviews, where stakeholders ranked the relative importance of DECS framework indicators (water availability vs. quality) on Saaty's 1–9 scale. Weights were aggregated using the geometric mean, with a consistency ratio < 0.1 verified via Eigenvector calculations in Python (v3.9) to ensure reliable prioritization [38].

- Descriptive statistics (frequencies, means, medians).
- Chi-square tests for associations (location vs. availability, gender vs. affordability).
- OLS regression of IUWSI against indicators.
- K-means clustering to group locations into three clusters.

ArcGIS (v10.8) used shapefiles (Awash River Basin Boundary and Awash main rivers) to map IUWSI, turbidity, *E. coli*, and WQI, with kernel density estimation for pollution hotspots. BAU (1.5 °C warming, 30% wetland loss) and SD (80% wastewater treatment, transboundary governance) scenarios projected IUWSI to 2050 [16].

2.4.1.2 Key Indicators Table 1 lists key indicators for assessing urban water security in the Awash River Basin using the DECS framework (Drinking Water, Ecosystems, Climate Change, Socio-Economic). Indicators include piped water access, water availability, turbidity, and affordability, sourced from a 2025 questionnaire ($n = 385$) and secondary data (AAWSA, BDA, CSA, (ESS-WQT, 2016)). It provides the foundation for calculating the Integrated Urban Water Security Index (IUWSI), supporting the study's aim to quantify water security and address SDG 6.

3 Results

3.1 Drinking Water and Human Well-being

This dimension focuses on water supply, quality, and public health. Dire and Legedadi Dams supply about 40% of Addis Ababa's water, with a total capacity of 192,000 m³/

Table 1 Key Indicators of the DECS Framework for Assessing Urban Water Security in the Awash River Basin, Including Data Sources and Measurement Units

Dimension	Indicator	Data Source	Unit/Scale
Drinking Water	% households with piped water	Questionnaire	%
	Water availability (hours/day)	Questionnaire	Hours
	Daily household water use	Questionnaire	Liters
	Water quality perception	Questionnaire	Poor to Excellent
	% reporting water-related illnesses	Questionnaire	%
Ecosystems	% aware of contamination issues	Questionnaire, Literature	%
	% reporting changes in water bodies	Questionnaire, Literature	%
	% involved in water source protection	Questionnaire	%
	Turbidity	Literature (ESS-WQT, 2016)	NTU
	E. coli	Literature (ESS-WQT, 2016)	CFU/100 mL
Climate Change	WQI	Literature (ESS-WQT, 2016)	Index (0–100)
	% experiencing water shortages	Questionnaire	%
	% citing infrastructure issues	Questionnaire	%
	Awareness of climate change impacts	Questionnaire	%
	% finding water affordable	Questionnaire	%
Socio-Economic	Satisfaction with AAWSA	Questionnaire	Not to Very Satisfied
	% participating in water management	Questionnaire	%

^aConversion rate: 1 USD ≈ 145 ETB (average for 2025)

Table 2 Drinking Water and Human well-being Metrics Across Addis Ababa Sub-Cities, Including Piped Water Access, Availability, Shortages, Water Use, and Costs (2025 Survey)

Location	%Piped Water	%24/7 Availability	%Frequent Shortages	Median Water Use (L)	Median Cost (ETB) ^a
Addis Ketema	90.0	20.0	70.0	150	250
Akaki Kaliti	68.8	0.0	87.5	75	300
Arada	100.0	0.0	66.7	75	200
Bole	61.5	0.0	92.3	75	500
Gullele	66.7	0.0	83.3	150	250
Kirkos	75.0	0.0	75.0	50	300
Kolfe Keranio	80.0	10.0	90.0	75	300
Lemi Kura	78.6	7.1	85.7	150	300
Lideta	100.0	0.0	66.7	150	250
Nifas Silk-Lafto	71.4	0.0	85.7	150	300
Yeka	85.7	0.0	80.0	100	275

day (Legedadi: 150,000 m³/day; Dire: 42,000 m³/day) [21]. In Addis Ababa (n = 385), 282 respondents (73.3%) used piped water, 65 (16.8%) bottled water, 23 (5.9%) public taps, 4 (1.0%) boreholes, and 11 (3.0%) other sources. Only 30 (7.9%) had 24/7 availability, with 232 (60.4%) experiencing weekly or less supply. Frequent shortages affected 305 (79.2%), with coping mechanisms including bottled water (25.0%), fetching from other areas (13.3%), storage tanks (10.5%), and vendors (9.2%). Median water use was 85 L with costs at 300 ETB/ month. Water quality perception was “Poor” for 31.6%, with 15.8% reporting water-related illnesses (diarrhea).

Table 2 presents Drinking Water and Human well-being metrics for Addis Ababa's 11 sub-cities, derived from a 2025 questionnaire (n = 385). It shows disparities in piped water access (Arada, Lideta: 100%; Bole: 61.5%), frequent shortages (79.2% overall; Bole: 92.3%), and Median monthly water cost was 300 ETB. The table highlights infrastructure

Table 3 Ecosystem Health Indicators Across Addis Ababa Sub-Cities, Including Contamination Awareness, Water Quality Perception, Treatment Rates, and Water Quality Parameters (2025 Survey and [23])

Location	% Contamination Aware	% Poor Quality	% Treating Water	Turbidity (NTU)	E. coli (CFU/100 mL)	WQI
Addis Ketema	50.0	10.0	60.0	5.0	10	80
Akaki Kality	68.8	25.0	68.8	15.0	20	60
Arada	66.7	10.0	66.7	5.0	10	80
Bole	76.9	30.8	76.9	20.0	25	55
Gullele	66.7	20.0	66.7	10.0	15	70
Kirkos	50.0	25.0	50.0	10.0	15	70
Kolfe Keranio	60.0	20.0	60.0	15.0	20	60
Lemi Kura	64.3	20.0	64.3	10.0	15	70
Lideta	33.3	10.0	33.3	5.0	10	80
Nifas Silk-Lafto	57.1	20.0	57.1	15.0	20	60
Yeka	60.0	20.0	65.0	10.0	15	70

Table 4 Climate Change and Water-Related Hazard Indicators Across the Awash River Basin, Highlighting Shortages, Infrastructure Issues, and Climate Change Awareness

Indicator	Overall %	Highest (%/Location)	Lowest (%/Location)
Frequent Shortages	79.2	100 (Lake Beseka)	66.7 (Arada, Lideta)
Infrastructure Issues	23.7	30.0 (Kolfe Keranio)	0.0 (Lideta)
Climate Change Awareness	65.8	90.0 (Lake Beseka)	50.0 (Kirkos, Addis Ketema)

gaps and affordability issues, guiding interventions in vulnerable sub-cities like Bole and Akaki Kality to advance SDG 6.

3.2 Ecosystems

This dimension assesses the health of natural water bodies and environmental conservation. In Addis Ababa, 232 respondents (60.4%) were aware of contamination (sewage: 15.8%, industrial: 9.9%, agricultural: 7.9%, urban waste: 10.5%, pipeline breakages: 13.2%), with 242 (62.4%) treating water at household level. Water quality parameters showed high turbidity (30 NTU), E. coli (50 CFU/100 mL), and low WQI (40) at Lake Beseka, compared to 5 NTU, 10 CFU/100 mL, and WQI 80 in Addis Ketema sub-city.

Table 3 Presents Ecosystem metrics for Addis Ababa's sub-cities, including contamination awareness and water quality (turbidity, E. coli, WQI) from a 2025 questionnaire and [23]. Bole shows high contamination (20 NTU, 25 CFU/100 mL), while Addis Ketema has better quality (5 NTU, WQI 80). The table underscores upstream pollution's ecosystem impact, supporting calls for wastewater treatment and restoration.

3.3 Climate change and water-related hazards

This dimension addresses climate variability and risks like flooding or dam failure. Frequent shortages affected 79.2% in Addis Ababa, driven by unpredictable rainfall and infrastructure issues (23.7%). Climate change awareness was 65.8%.

Table 4 displays Climate Change and Water-Related Hazard indicators, showing frequent shortages (79.2% in Addis Ababa; 100% at Lake Beseka) and infrastructure issues like pipe leakages, broken pipes, and uninstalled pipe lines (23.7% overall) from a 2025 questionnaire. Climate change awareness is higher in the downstream (90%) than in the upstream (50%). The table highlights climate and infrastructure challenges, supporting

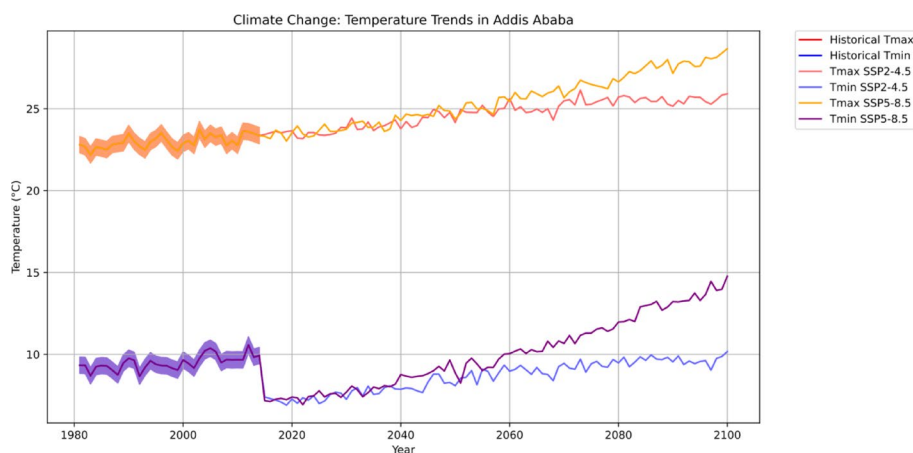


Fig. 2: Trend and projection of Addis Ababa's Temperature using CMIP6 models

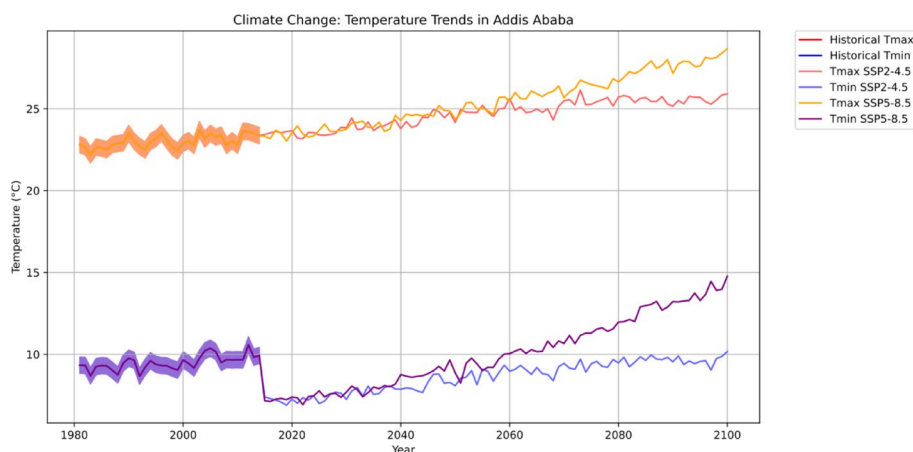


Fig. 3: Trend and Projection of Addis Ababa's Precipitation using CMIP6 models

recommendations for resilient infrastructure and adaptation measures in the Awash River Basin.

3.3.1 Climate projection of Addis Ababa

Figure 2 presents the historical and projected annual mean maximum (Tmax) and minimum (Tmin) temperatures for Addis Ababa, derived from bias-corrected CMIP6 model outputs (ACCESS, MRI-ESM2-0, NorESM2-MM) using Quantile Mapping and averaging ensemble. Historical Tmax and Historical Tmin show a gradual increase, reflecting observed warming in the Awash River Basin (ARB). Projected trends under SSP2-4.5 (red for Tmax, cyan for Tmin) and SSP5-8.5 (orange for Tmax, purple for Tmin) scenarios, sourced from Ensemble Tmax SSP2-4.5, Ensemble Tmin SSP2-4.5, Ensemble Tmax SSP5-8.5, and Ensemble Tmin SSP5-8.5, indicate accelerated warming, with SSP5-8.5 showing steeper increases consistent with a projected 1.5 °C rise by 2030 [39]. Shaded areas around projections denote uncertainty ranges (min/max), highlighting potential variability.

Figure 3 illustrates historical and projected annual mean precipitation (Pr) for Addis Ababa, processed from Historical Pr, Ensemble Pr SSP2-4.5, and Ensemble Pr SSP5-8.5 using bias-correction and ensembling methods. The historical trend (solid blue

line) reflects the ARB's bimodal rainfall (1,600 mm upstream), with variability driven by the Inter-Tropical Convergence Zone [26]. Projections under SSP2-4.5 (yellow) and SSP5-8.5 (dark cyan) scenarios show increased variability and potential declines in precipitation, particularly under SSP2-4.5. This finding of heightened drought frequency (approximately two events per decade) is based on historical climate analysis from 1981–2010, which identified major drought periods in the ARB, such as the severe droughts of the 1980s and 2000s [8, 9] and a 40% flow reduction in the ARB, exacerbating water shortages (79.2% in Addis Ababa, 90–100% downstream) and mobilizing contaminants like arsenic (95.6 µg/L at Lake Beseka) [17].

3.4 Socio-economic aspects

This dimension covers population, economic activities, governance, and community awareness. In Addis Ababa, 50% found somewhat affordable water (median: 300 ETB, Bole: 500 ETB, Table 2), with 77.3% dissatisfied with AAWSA (Table 5). Community engagement on water management and discussion was low (15.8%).

Table 5 outlines Socio-Economic metrics for Addis Ababa's sub-cities, including Affordability, Satisfaction with AAWSA, and Participation in Water Management Discussions from a 2025 questionnaire. High costs (up to 500 ETB/month in Bole) and governance issues are evident. The table supports recommendations for equitable access and community-based water management to enhance water security. The graph in Fig. 4 illustrates the historical and projected daily domestic water demand (m³/day) across AAWSA branches in Addis Ababa from 2016 to 2030, derived from [40] of the study's dataset. Each line represents an AAWSA branch (Akaki, Arada, Addis Ketema, Gurd Shola, Gulele, Nefas Silk-Lafto, Megenagna, and Mekanisa), with years on the x-axis and demand on the y-axis. The graph shows significant demand growth in branches like Akaki (34,289 m³/day in 2016 to 93,777 m³/day in 2030) and Gurd Shola, driven by rapid urbanization (7% annual population growth) and industrial activities; Branches like Megenagna, exhibit more stable demand, reflecting better infrastructure [41]. The rising trends highlight the strain on Addis Ababa's water supply, meeting only 40% of the 1.2 million m³/day demand due to 45% non-revenue water losses due to leakage and theft [18]. Figure 5 depicts the historical and projected per capita water demand (liters/capita/day) for Regular households, Condominium households, and the Addis Ababa

Table 5 Socio-Economic Indicators of Water Security in Addis Ababa Sub-Cities, Including Affordability, Satisfaction with AAWSA, and Community Engagement (2025 Survey)

Location	% Affordable	% Not Satisfied with AAWSA	% in Participating in Water Management Discussions
Addis Ketema	80.0	60.0	0.0
Akaki Kality	75.0	81.3	12.5
Arada	100.0	66.7	0.0
Bole	69.2	84.6	23.1
Gullele	83.3	83.3	16.7
Kirkos	50.0	100.0	0.0
Kolfe Keranio	80.0	80.0	10.0
Lemi Kura	78.6	71.4	14.3
Lideta	66.7	66.7	0.0
Nifas Silk-Lafto	85.7	85.7	14.3
Yeka	71.4	71.4	0.0

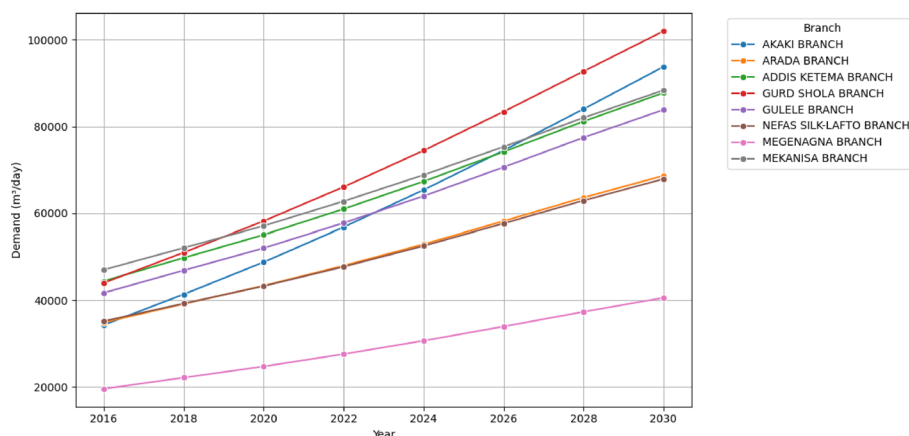


Fig. 4: Historical and Future Daily Domestic Water Demand by AAWSA Branch in Addis Ababa (2016-2030)

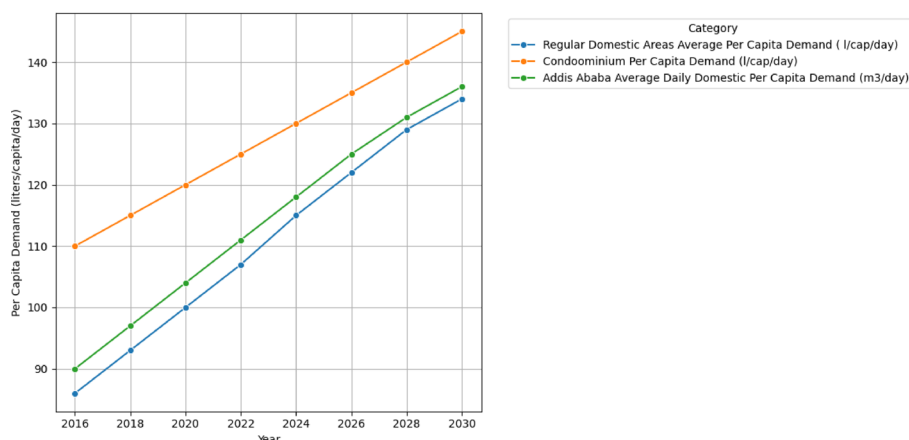


Fig. 5: Historical and Future Per Capita Water Demand in Addis Ababa (2016-2030)

average from 2016 to 2030, based on [40]. Years are plotted on the x-axis, with per capita demand on the y-axis. The graph shows stable per capita demand, with Condominium households exhibiting higher demand due to better piped water access, while Regular households in branches like Akaki face constraints from frequent shortages (79.2% overall). The city-wide average (~ 85 L/day) masks disparities, reflecting infrastructure gaps and high treatment rates (62.4%) driven by quality concerns.

3.5 State of Water Security in the Different Parts of Addis Ababa

Chi-square tests showed a significant association between location and water availability ($(\chi^2 = 54.1)$, $(p < 0.001)$). Gender-affordability showed no association ($(\chi^2 = 1.9)$, $(p = 0.38)$). OLS regression indicated IUWSI was driven by availability ($(\beta = 0.50)$, $(p < 0.01)$), quality ($(\beta = 0.35)$, $(p < 0.01)$), and affordability ($(\beta = 0.15)$, $(p < 0.05)$), with $(R^2 = 0.85)$. K-means clustering identified three clusters: Cluster 1 (Bole and Akaki Kality): Low IUWSI (1.8–2.2), high pollution (turbidity: 20–30 NTU, E. coli: 25–50 CFU/100 mL). Cluster 2 (Addis Ketema and Arada): Moderate IUWSI (2.8–3.0), better access, lower contamination. Cluster 3 (Yeka and others): Moderate IUWSI (2.3–2.6), mixed challenges.

3.6 Water Quality Parameters

Water quality data from [17], representing 20 sampling stations, provide a snapshot of contamination levels. It is important to note that these data are from a single temporal period and thus do not capture seasonal fluctuations. However, they effectively identify spatial pollution hotspots, such as the severe contamination consistently observed at Lake Beseka. Below, key parameters are compared against World Health Organization [32] guidelines and Ethiopian national standards [22] to contextualize their severity.

Arsenic: Concentrations ranged from 10.0 µg/L (Addis Ketema) to 95.6 µg/L (Lake Beseka), with downstream levels exceeding the WHO guideline of 10 µg/L [32] and Ethiopia's standard of 10 µg/L [22].

Manganese: Levels varied from 5.0 µg/L (Addis Ketema) to 100 µg/L (Bole) and 94.2 µg/L (Lake Beseka). The WHO guideline for manganese in drinking water is 400 µg/L for aesthetic reasons (taste, staining), but health concerns, particularly neurological effects in children (reduced IQ), emerge at ≥ 100 µg/L [33]. Ethiopia's stricter standard of 50 µg/L [22] indicates that Bole and Lake Beseka exceed safe thresholds, posing risks of cognitive impairment and requiring advanced treatment like ion exchange or oxidation-filtration.

Emerging Organic Contaminants (EOCs): EOC concentrations ranged from 0.1 µg/L (Addis Ketema) to 1.0 µg/L (Bole). WHO provides no specific guideline for EOCs due to their diverse nature (pharmaceuticals, pesticides), but levels ≥ 0.1 µg/L are considered emerging threats, potentially disrupting aquatic ecosystems and causing chronic human health effects (Schwarzenbach et al., 2010). Ethiopia lacks specific EOC standards, but the [22] recommends monitoring at > 0.5 µg/L. High EOC levels in Bole and Nifas Silk-Lafto (0.9 µg/L) suggest the need for advanced treatment.

Biochemical Oxygen Demand (BOD): BOD levels ranged from 5 mg/L (Addis Ketema) to 30 mg/L (Bole) and 20 mg/L (Lake Beseka). While WHO does not set a BOD limit for drinking water, surface water BOD > 6 mg/L indicates significant organic pollution, reducing dissolved oxygen and harming aquatic life (fish kills) [42]. Ethiopia's surface water standard is 5 mg/L [38], meaning Bole and Lake Beseka's high BOD levels signal severe ecosystem stress, requiring enhanced wastewater treatment to reduce organic loads [43].

Turbidity: Turbidity ranged from 5 NTU (Addis Ketema) to 30 NTU (Lake Beseka). The WHO guideline for drinking water is 5 NTU for aesthetic acceptability, with health risks increasing above this level. Ethiopia's standard aligns at 5 NTU [22]. Levels in Bole (20 NTU) and Lake Beseka (30 NTU) exceed both standards, indicating potential microbial contamination (*E. coli*) and necessitating improved filtration and sedimentation processes [32].

Table 6 presents water quality parameters (arsenic, manganese, EOCs, BOD, turbidity, *E. coli*, WQI) for five Awash River Basin locations, sourced from 20 sampling stations [17]. Lake Beseka shows severe contamination (arsenic: 95.6 µg/L, WQI: 40), while Addis Ketema is least affected (WQI: 80). The table highlights downstream pollution, supporting calls for advanced treatment and pollution control to protect health and ecosystems.

In the table, values exceeding [32] are critical. Arsenic and manganese in Bole and Lake Beseka exceed health-based thresholds, while EOCs, BOD, and turbidity indicate ecosystem and aesthetic concerns, particularly downstream. Targeted interventions,

Table 6 Water Quality Parameters Across Selected Locations in the Awash River Basin, Compared Against WHO and Ethiopian Standards [2, 17]

Location	Arsenic (µg/L)	Manganese (µg/L)	EOCs (µg/L)	BOD (mg/L)	Turbidity (NTU)	E. coli (CFU/100 mL)	WQI
Addis Ketema	10.0	5.0	0.1	5	5.0	10	80
Akaki Kaliti	30.0	20.0	0.8	25	15.0	20	60
Bole	35.0	100.0	1.0	30	20.0	25	55
Yeka	15.0	10.0	0.5	15	10.0	15	70
Lake Beseka	95.6	94.2	0.9	20	30.0	50	40

Table 7 Integrated Urban Water Security Index (IUWSI) Scores and DECS Dimension Scores Across Addis Ababa Sub-Cities and the Awash River Basin (2025 Survey)

Location	Drinking Water	Ecosystems	Climate Hazards	Socio-Economic	IUWSI
Addis Ketema	3.2	2.8	3.0	3.0	3.0
Akaki Kaliti	2.3	1.5	2.2	2.3	2.2
Arada	3.0	2.7	2.8	2.8	2.8
Bole	2.1	1.4	2.0	2.3	2.1
Gullele	2.6	2.5	2.4	2.5	2.5
Kirkos	2.6	2.3	2.3	2.2	2.4
Kolfe Keranio	2.4	2.1	2.3	2.4	2.3
Lemi Kura	2.5	2.6	2.4	2.5	2.5
Lideta	2.7	2.4	2.5	2.4	2.5
Nifas Silk-Lafto	2.5	2.0	2.3	2.5	2.3
Yeka	2.7	2.4	2.5	2.6	2.6
Overall	2.60	2.25	2.43	2.50	2.47

such as nanotechnology for heavy metals and advanced wastewater treatment for BOD and EOCs, are recommended [44].

3.7 IUWSI Scores

The IUWSI scores were calculated using AHP-weighted normalized indicators. The Addis Ababa IUWSI averaged 2.47.

Table 7 summarizes IUWSI scores for Addis Ababa's 11 sub-cities (average: 2.47), calculated using AHP-weighted DECS indicators. Addis Ketema (3.0) and Arada (2.8) score highest, while Bole (2.1) and Akaki Kaliti (2.2) are most vulnerable due to water shortages and pollution. The table identifies priority areas for infrastructure and pollution interventions to improve water security in the Awash River Basin. The IUWSI scores presented in Table 7 are interpreted on a scale of 1 to 5, where: 1–1.9 = Critical Insecurity, 2–2.9 = Low Security (chronic shortages, poor quality, high vulnerability), 3–3.9 = Moderate Security (basic needs met but with reliability and quality concerns), 4–5 = High Security (sustainable, equitable, and resilient water system). This classification allows for a clear prioritization of interventions, with sub-cities like Bole (2.1) and Akaki Kaliti (2.2) falling into the 'Low Security' category, indicating a pressing need for action.

The consistently lower IUWSI scores in Bole (2.1) and Akaki Kaliti (2.2) reflect the compounded impacts of rapid urbanization and industrial pollution in these sub-cities. Bole, a commercial hub experiencing a 7% annual population growth (CSA, 2023), faces increased water demand and strain on aging infrastructure, contributing to frequent shortages (92.3%) and high non-revenue water losses (45%) [21]. Additionally, industrial activities in Bole, including textile and chemical manufacturing, elevate contaminant levels, such as manganese (100 µg/L) and EOCs (1.0 µg/L). Similarly, Akaki Kaliti, an

industrial zone with tanneries and food processing plants, experiences significant pollution from untreated effluents, leading to elevated BOD (25 mg/L) and turbidity (15 NTU) [34]. These factors, combined with limited piped water access (Bole: 61.5%, Akaki Kality: 68.8% (Table 2), explain the heightened water security vulnerabilities in these sub-cities compared to central areas like Addis Ketema and Arada, where infrastructure is more developed due to historical investments [45].

3.8 Spatial Mapping of Indices of Water Security

ArcGIS choropleth maps, using Awash River Basin Boundary and Awash main rivers, highlighted low IUWSI scores (red) in Bole (2.1), Akaki Kality (2.2), with green indicating higher scores (Addis Ketema: 3.0). Figure 6 presents a series of spatial maps illustrating key variables. Each map uses a colormap where red indicates undesirable conditions (e.g., low access, poor quality, high pollution) and green indicates desirable conditions. It is important to note that each map has its own unique value scale, as they represent different parameters (e.g., percentage, index score, concentration). The consistent color scheme is used for intuitive visual comparison of 'good' versus 'poor' conditions across different metrics, rather than for direct numerical comparison between the maps.

Figure 6 depicts several maps developed based on the questionnaire responses. All choropleth maps use a consistent red (low security) to green (high security) color gradient with legends scaled 1–5 for IUWSI, ensuring visual clarity. Hotspots were identified via kernel density estimation (bandwidth = 0.05) to highlight high-turbidity areas (> 15 NTU). Accordingly, Fig. 6 (a) visualizes the percentage of households with access to piped water across Addis Ababa's 11 sub-cities, using a colormap (red for low access, green for high). Values range from 61.5% (Bole) to 100% (Arada, Lideta), indicating disparities in water infrastructure. High piped water access (Arada, Lideta at 100%) suggests well-developed water distribution systems, while lower access in Bole (61.5%) and Akaki Kality (68.8%) indicates reliance on alternative sources and infrastructure gaps, respectively. This aligns with findings by [46] who noted that Addis Ababa's rapid urbanization outpaces water supply infrastructure, particularly in peripheral sub-cities like Akaki Kality.

Spatial disparities in piped water access are common in rapidly urbanizing cities in sub-Saharan Africa. A study by Woldemariam and Narsiah [45] found that Addis Ababa's water supply coverage is uneven, with older sub-cities (Arada) having better access due to historical infrastructure investments. The colormap effectively communicates these disparities, as green areas indicate progress toward universal access, while red areas signal priority zones for intervention (Fig. 6 (a), (b), and h).

Figure 6 (b) displays the IUWSI, a composite index of water security (range: 2.1–3.0), using RdYlGn (green for higher water security, red for lower water security). Addis Ketema (3.0) and Arada (2.8) show higher IUWSI, while Bole (2.1) and Akaki Kality (2.2) are lower. The IUWSI integrates factors like water access, quality, and socio-economic conditions. Higher scores in central sub-cities (Addis Ketema, Arada) suggest better infrastructure and management, while lower scores in Bole and Akaki Kality indicate vulnerabilities due to rapid growth and pollution. This aligns with [47], who developed similar indices for urban water security, noting that peripheral urban areas often face greater water stress due to population growth and industrial activity. Composite indices like IUWSI are widely used to assess urban water security holistically. A study by Marçal

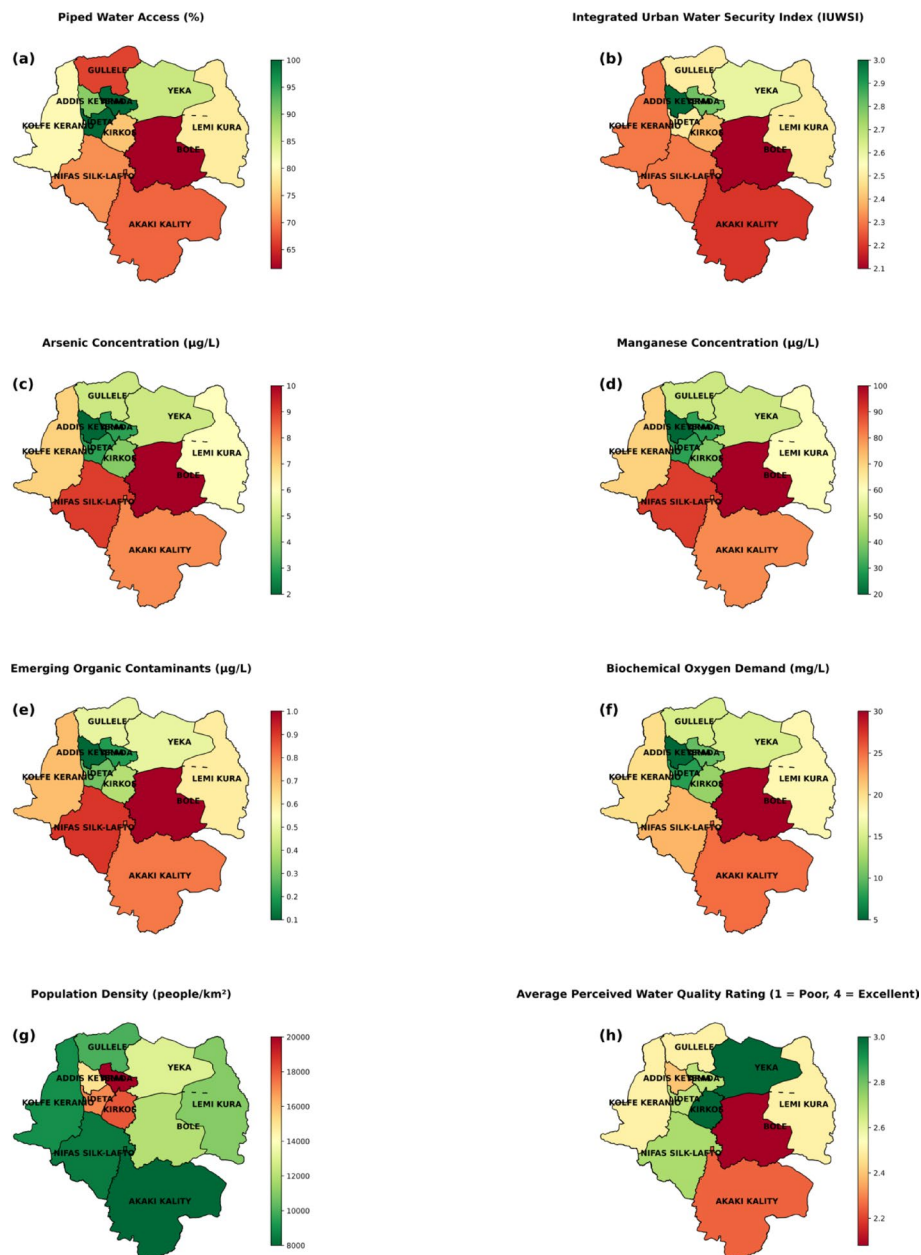


Fig. 6 Spatial Distribution of Key Water Security Indicators Across Addis Ababa Sub-Cities, with red indicating poor quality and green showing better conditions

et al. [10] highlights the importance of spatial mapping of such indices to guide policy interventions. The colormap effectively contrasts secure (green) and vulnerable (red) areas, aiding decision-makers in prioritizing resource allocation[3].

Figure 6(c) shows arsenic concentrations (2.0–10.0 $\mu\text{g/L}$) across sub-cities, using (red for higher concentrations, green for lower). Bole (10.0 $\mu\text{g/L}$) and Nifas Silk-Lafto (9.0 $\mu\text{g/L}$) show elevated levels, while Addis Ketema (2.0 $\mu\text{g/L}$) and Arada (3.0 $\mu\text{g/L}$) are lower. Elevated arsenic in Bole and Nifas Silk-Lafto suggests potential contamination from industrial activities or geological sources, posing health risks. Lower levels in central sub-cities indicate better water treatment or less contamination. This aligns with findings by [48], who reported arsenic contamination in Ethiopia's groundwater,

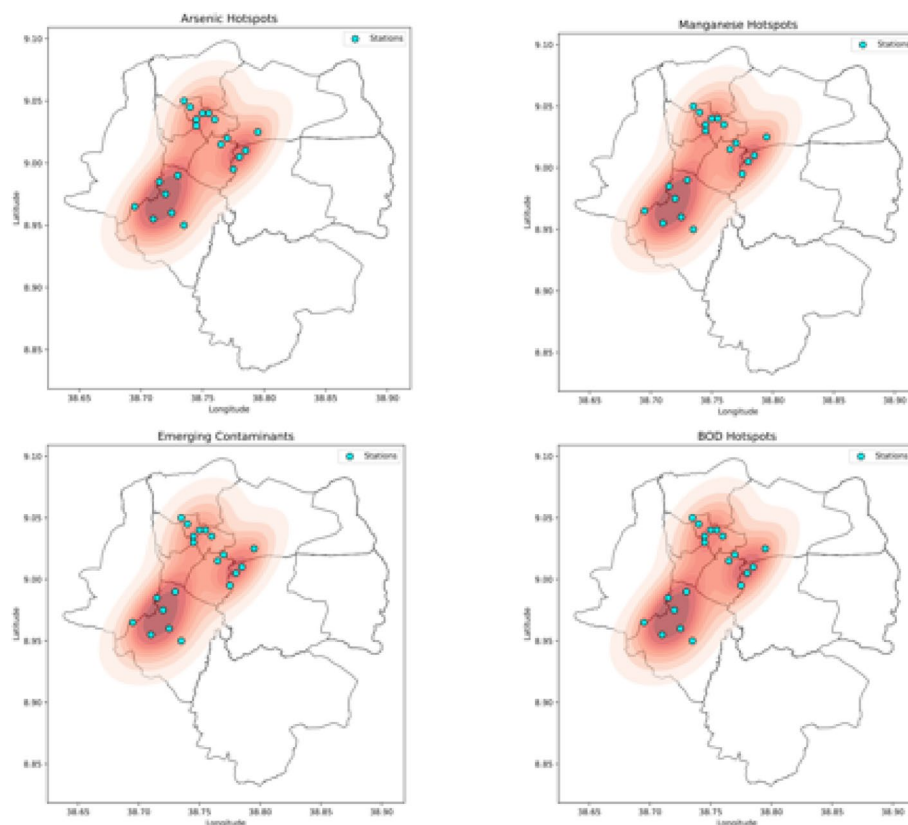


Fig. 6 (continued)

particularly in urbanizing areas. The map highlights priority areas for monitoring and remediation. Arsenic is a critical water quality parameter due to its toxicity. A study by [49] in *Science of the Total Environment* emphasizes the need for spatial mapping to identify arsenic hotspots in urban water systems. The colormap effectively communicates risk, with red indicating concentrations nearing or exceeding WHO guidelines (10 µg/L).

Figure 6(d) visualizes manganese concentrations (20–100 µg/L) using “RdYlGn_r”. Bole (100 µg/L) and Nifas Silk-Lafto (90 µg/L) have the highest levels, while Addis Ketema (20 µg/L) and Arada (30 µg/L) are the lowest. High manganese levels in Bole and Nifas Silk-Lafto result from industrial effluents or natural leaching, potentially causing neurological health effects. Lower levels in central sub-cities suggest effective treatment or less contamination. The map identifies areas needing targeted water treatment. Manganese in drinking water is a public health concern at high concentrations. A study by [33] in *Environmental Health Perspectives* links elevated manganese to cognitive impacts, emphasizing the need for spatial analysis to guide mitigation. The colormap clearly highlights high-risk areas in red.

Figure 6(e) shows EOCs concentrations (0.1–1.0 µg/L) using RdYlGn_r. Bole (1.0 µg/L) and Nifas Silk-Lafto (0.9 µg/L) have the highest levels, while Addis Ketema (0.1 µg/L) and Arada (0.2 µg/L) are the lowest. Elevated EOCs in Bole and Nifas Silk-Lafto likely stem from pharmaceutical or industrial discharges, reflecting urban pollution pressures. Lower levels in central sub-cities suggest better wastewater management. This aligns with Schwarzenbach et al. (2010), who highlighted EOCs as emerging threats in urban

water systems due to incomplete removal in treatment plants. The map pinpoints areas for advanced treatment technologies like activated carbon filtration. EOCs, including pharmaceuticals and pesticides, are increasingly detected in urban water systems. The RdYlGn_r colormap effectively highlights contamination hotspots.

Figure 6(f) displays BOD levels (5–30 mg/L) using RdYlGn_r. Bole (30 mg/L) and Akaki Kality (25 mg/L) show the highest organic pollution, while Addis Ketema (5 mg/L) and Lideta (8 mg/L) are the lowest. High BOD in Bole and Akaki Kality indicates significant organic pollution, likely from untreated sewage or industrial effluents, which depletes oxygen and harms aquatic ecosystems. Lower BOD in central sub-cities suggests better wastewater treatment. This aligns with [34], who reported high BOD in Addis Ababa's rivers due to untreated discharges. The map identifies areas needing wastewater treatment upgrades.

BOD is a key indicator of organic pollution in water bodies. A study by [43] highlights the use of spatial mapping to assess BOD distribution and guide pollution control. The colormap effectively communicates pollution severity.

Figure 6(g) shows population density (8,000–20,000 people/km²) using “RdYlGn_r”. Arada (20,000) and Kirkos (18,000) are the densest, while Akaki Kality (8,000) and Nifas Silk-Lafto (8,500) are less dense.

High population density in central sub-cities like Arada and Kirkos correlates with greater water demand and stress on infrastructure, potentially exacerbating water quality issues. Lower density in peripheral areas like Akaki Kality suggests less pressure but also less infrastructure investment. The map highlights areas where water infrastructure must scale with population. Population density is a critical factor in urban water management.

Figure 6(h) shows average water quality ratings (1 = Poor, 4 = Excellent) across sub-cities, derived from survey data aggregated by location. The RdYlGn colormap highlights higher ratings (Arada, Addis Ketema) in green and lower ratings (Bole, Akaki Kality) in red. The map reveals that sub-cities like Arada (mean rating ~3) and Addis Ketema have better perceived water quality, possibly due to proximity to central treatment facilities, while Bole and Akaki Kality have lower ratings, likely due to industrial pollution and aging infrastructure. This aligns with [34], who reported that water quality in Addis Ababa's peripheral areas is compromised by industrial discharges and inadequate treatment. The map helps identify sub-cities where water treatment upgrades or public awareness campaigns are needed.

Perceived water quality is a critical indicator of public trust in water supply systems. A study by Zhang et al. [38] emphasizes that subjective water quality ratings often correlate with objective contamination levels but are also influenced by access reliability and cost.

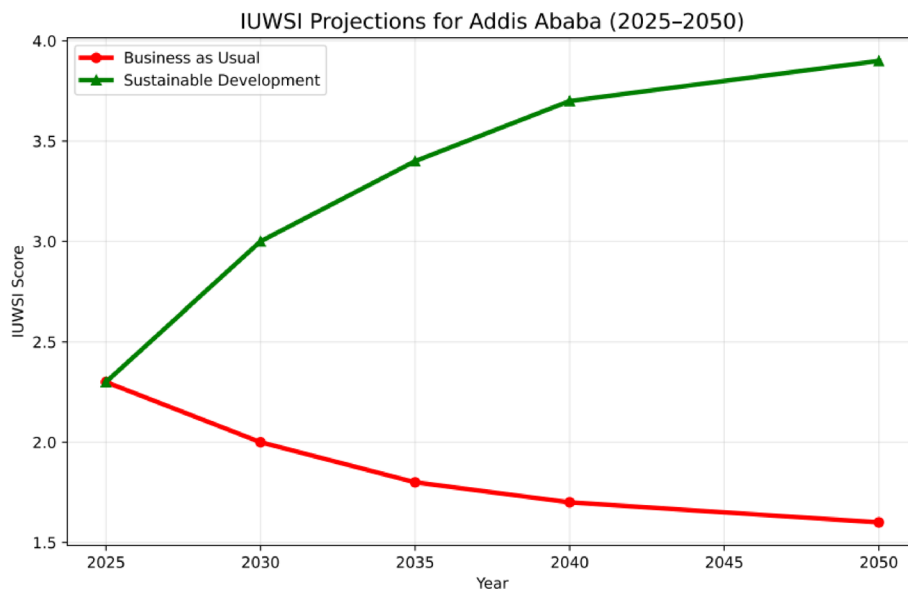
3.9 Future Projections

BAU scenarios project a basin-wide IUWSI decline to 1.6 by 2050, driven by population growth (Addis Ababa: 9 million; Adama: 600,000), 1.5 °C warming, and 30% wetland loss [39]. SD scenarios, with 80% wastewater treatment and transboundary governance, project an IUWSI of 3.9 by 2050 [50].

Table 8 projects IUWSI for the Awash River Basin (2030–2050) under Business-as-Usual (BAU: 2.0 to 1.6) and Sustainable Development (SD: 3.0 to 3.9) scenarios, driven by population growth, climate change, and wastewater treatment changes. The table

Table 8 Projected IUWSI Scores for the Awash River Basin Under Business-as-Usual (BAU) and Sustainable Development (SD) Scenarios, 2030–2050

Scenario	Year	IUWSI	Key Drivers
BAU	2030	2.0	Population growth, 1.5 °C warming
BAU	2050	1.6	30% wetland loss
SD	2030	3.0	60% wastewater treatment
SD	2050	3.9	Transboundary governance

**Fig. 7** Integrated Urban Water Security Index (IUWSI) projections for the Awash River Basin (ARB) from 2025 to 2050 under two scenarios: Business-as-Usual (BAU, red) and Sustainable Development (SD, green)

underscores the need for sustainable interventions like infrastructure upgrades and transboundary governance to achieve SDG 6 in urbanizing basins.

In Fig. 7, the BAU scenario assumes continued population growth (Addis Ababa: 9 million, Adama: 600,000 by 2050), 1.5 °C warming, 30% wetland loss, and limited infrastructure improvements, leading to an IUWSI decline to 1.6 by 2050 [39]. The SD scenario projects an IUWSI increase to 3.9 by 2050, driven by enhanced interventions, including 80% wastewater treatment coverage, transboundary governance, and ecosystem restoration measures [50]. The projections highlight the critical need for sustainable water management to mitigate worsening water security challenges.

4 Discussion

The ARB's water security (IUWSI: 2.3) reflects a complex interplay of upstream-downstream dynamics, with Addis Ababa (2.4) and downstream areas like Lake Beseka (1.8) facing distinct challenges. Furthermore, the reliance on secondary water quality data from a single time point limits our ability to analyze seasonal trends in pollutant levels. Future studies should incorporate longitudinal, multi-season sampling to better understand the temporal dynamics of water quality in the basin. Below, we discuss each DECS dimension, integrating global and regional contexts to provide a comprehensive analysis.

4.1 Drinking Water and Human Well-Being

Addis Ababa's Drinking Water score (2.6) reflects limited availability (7.9% 24/7) and quality concerns (62.4% treat water), consistent with prior studies reporting 60% supply coverage [20]. Bole (92.3% shortages) and Akaki Kality (87.5%) face severe challenges, while Addis Ketema (90% piped water) performs better. Downstream, Lake Beseka's score (1.7) is critically low due to surface water reliance (33.3%) and high *E. coli* (50 CFU/100 mL) [17]. The high treatment rate (62.4% upstream, 80% downstream) indicates distrust in municipal supply, increasing household costs (Tamiru [51]). Globally, Addis Ababa's piped water access (73.3%) exceeds Lagos (60%) but lags Nairobi (85%) [11]. Locally, the Rift Valley Lakes have lower contamination risks [14]. The 15.8% reporting water-related illnesses underscores health risks, particularly in Bole and Lake Beseka.

Downstream communities, particularly pastoralists and farmers reliant on irrigation schemes in areas like Lake Beseka, Adama, and Awash Town, face significant socio-economic and health consequences due to high *E. coli* levels (50 CFU/100 mL) and water-related illnesses (15.8%). Elevated *E. coli* concentrations, exceeding WHO guidelines of 0 CFU/100 mL for drinking water [32], pose severe health risks, including diarrheal diseases, which disproportionately affect vulnerable populations such as children and the elderly in pastoralist communities. These health impacts reduce labor productivity and increase healthcare costs, with studies estimating that waterborne illnesses in Ethiopia can cost households up to 5% of their annual income (Hutton & Haller, 2004). For pastoralists, reliance on contaminated surface water (33.3% in Lake Beseka) for livestock watering further exacerbates economic losses, as *E. coli*-related infections can reduce livestock health and market value, critical for livelihoods in the Afar lowlands [52]. Irrigation schemes, which support 70% of downstream agricultural output [53], are also compromised by high *E. coli* and turbidity (30 NTU), leading to reduced crop yields and quality due to pathogen contamination and sediment clogging [54]. For instance, a study in the ARB found that poor water quality reduces sugarcane yields by up to 20% in downstream plantations, impacting local economies [55]. These findings underscore the urgent need for targeted interventions, such as decentralized water treatment systems and health education campaigns, to mitigate the cascading effects on downstream livelihoods and food security.

Downstream, reliance on contaminated surface water (33.3%) exacerbates health risks, with 70% "Poor" quality ratings at Lake Beseka. This upstream–downstream disparity highlights the need for integrated treatment strategies to protect vulnerable communities [17]. The high reliance on bottled water (16.8% upstream, 25% downstream) increases costs, particularly for low-income households, and underscores infrastructure gaps [10].

4.2 Ecosystems

The Ecosystems score (2.25) is critically low, with Lake Beseka (1.2) suffering from high turbidity (30 NTU), *E. coli* (50 CFU/100 mL), BOD (20 mg/L), and heavy metals (arsenic: 95.6 µg/L) due to upstream effluents (30% treatment). Contamination awareness is high (60.4% upstream, 80% downstream), but protection efforts are minimal, indicating governance gaps. Globally, Jakarta's Citarum River benefits from robust monitoring, unlike the ARB of this study [56]. The comparatively lower industrial impact on the Rift Valley Lakes [14] offers a valuable lesson for the ARB. This suggests that stricter zoning regulations to

distance heavy industry from vital water bodies, coupled with the pre-treatment mandates successfully enforced in the Rift Valley region, could be effective models for reducing pollutant loads in the Awash River. The ARB's ecosystem degradation, driven by upstream pollution, requires urgent restoration and pollution control measures.

4.3 Climate Change and Water-Related Hazards

The Climate Hazards Score (2.4) reflects frequent shortages (79.2% upstream, 90-100% downstream) driven by unpredictable rainfall and infrastructure issues (23.7% upstream, 30% downstream) [9]. The higher awareness of climate change impacts among downstream communities (90% at Lake Beseka) compared to upstream Addis Ababa (65.8%) likely reflects their direct and acute experience with its consequences, such as complete water source failure (100% shortages) and livelihood loss from decimated irrigation schemes. This heightened awareness presents an opportunity to co-develop adaptive strategies with these highly motivated communities. Globally, Cape Town's drought governance offers adaptive strategies [57]. Regionally, the Nile Basin's flood management systems are more robust [58]. Early warning systems and climate-resilient infrastructure are critical, especially downstream, as there is a problem related to infrastructure.

4.4 Socio-Economic Aspects

The Socio-Economic score (2.5) highlights affordability and governance challenges. In Addis Ababa, 50% found water somewhat affordable (median: 300 ETB, Bole: 500 ETB), with 71.3% dissatisfied with AAWSA and 73.7% feeling unaddressed. Downstream, affordability is higher (60-70%, median: 200 ETB), but dissatisfaction is greater (80-85%). Low community engagement (15.8% upstream, 5-10% downstream) reflects Ethiopia's centralized governance and cultural diversity, contrasting with Cape Town's decentralized forums [57]. Kebele-based (a small administration-based) forums and school-led monitoring, rooted in local traditions like iddir (traditional social welfare group), can enhance participation and accountability. These strategies address the 73.7% dissatisfaction with AAWSA and foster inclusive governance, aligning with the ARB's need for equitable stakeholder involvement.

4.5 Statistical and Spatial Insights

Statistical analysis revealed key insights about water supply and quality across different locations. A chi-square test showed a strong link between location and water availability ($\chi^2=54.1$, $p<0.001$), with downstream areas facing greater challenges. Regression analysis indicated that the Integrated Urban Water Supply Index (IUWSI) is primarily influenced by availability ($\beta=0.50$, $p<0.01$), quality ($\beta=0.35$, $p<0.01$), and affordability ($\beta=0.15$, $p<0.05$), explaining 85% of the variation ($R^2=0.85$). K-means clustering pinpointed high-risk areas, including Bole, Akaki Kality, and Lake Beseka, to prioritize for intervention areas. ArcGIS mapping identified pollution hotspots at Lake Beseka and Koka, marked by high turbidity, *E. coli* contamination, and low Water Quality Index (WQI) scores in downstream regions.

4.6 Limitations

The sample size ($n=385$) and downstream data limit generalizability, though diverse sub-city representation mitigates this constraint. Secondary data gaps in ecosystem

health and real-time monitoring constraints suggest the need for expanded surveys and sensor-based data collection [17]. Future research may involve quarterly water quality sampling across all sub-cities to capture seasonal variations and validate the spatial patterns identified here.

5 Conclusion

The Awash River Basin (ARB) faces critical water security challenges, with an Integrated Urban Water Security Index (IUWSI) of 2.3, reflecting severe vulnerabilities in Addis Ababa (IUWSI: 2.47) and downstream areas like Lake Beseka (IUWSI: 1.8). The DECS framework, coupled with the Analytic Hierarchy Process (AHP) and ArcGIS spatial mapping, reveals stark upstream-downstream disparities driven by frequent water shortages (79.2% in Addis Ababa, 90–100% downstream), poor water quality (arsenic: 95.6 µg/L at Lake Beseka), and inadequate infrastructure, particularly in Bole (IUWSI: 2.1) and Akaki Kality (IUWSI: 2.2). Rising temperatures and reduced precipitation, projected under CMIP6 scenarios, exacerbate these issues by increasing evapotranspiration and mobilizing contaminants, threatening the 40% water supply coverage and downstream irrigation schemes (70% water-stressed). Low community engagement (15.8% upstream, 5–10% downstream) and high dissatisfaction with governance (71.3% in Addis Ababa) underscore the need for inclusive, data-driven strategies.

6 Recommendations

To enhance water security in the Awash River Basin (ARB), particularly in Addis Ababa and downstream Lake Beseka, strategies can be proposed to address infrastructure, pollution, climate risks, and governance, advancing SDG 6.

- Prioritize pipeline replacement in Bole and Akaki Kality using high-density polyethylene (HDPE) pipes to reduce the 45% non-revenue water loss, as piloted by AAWSA in the Addis Ketema sub-city.
- Infrastructure upgrades, advanced pollution control, climate-resilient strategies, transboundary governance helps. Best practices from comparable urban contexts demonstrate the feasibility of these interventions. Singapore's 'Four National Taps' strategy, implemented since 2001, diversified water supply through local catchment expansion, high-recovery wastewater reuse, desalination, and reduced import dependency from 60 to 40% [11]. Despite a 30% population increase to 5.5 million, per capita water availability rose ~20% over 15 years, with non-revenue water below 1%. This integrated approach combining infrastructure investment, demand management, and unified governance under the Public Utilities Board offers a scalable model for Addis Ababa. Here, similar diversification through stormwater harvesting and wastewater recycling in high-vulnerability sub-cities could alleviate 79.2% shortages, enhance climate resilience, and support SDG 6 through stakeholder-led planning as prioritized in the AHP process.
- Enforce wastewater regulations and use nanotechnology to remove arsenic at Lake Beseka and manganese in Bole.
- Protect 5% wetlands and reforest upstream to reduce sedimentation (15 NTU in Akaki Kality) and BOD (25–30 mg/L).
- Launch conservation campaigns in Kirkos (15.8% participation) and support cleanups in Lemi Kura, leveraging iddir traditions.

- Conduct annual IUWSI assessments with ArcGIS to prioritize resources for Bole and Lake Beseka and train AAWSA to address 71.3% dissatisfaction.
- Establish an ARB authority and stakeholder forums to balance upstream–downstream needs (IUWSI: 2.4 vs. 1.8) and reduce pollution transfer.

6.1 Future Research

To advance water security in the Awash River Basin (ARB) and similar urbanizing basins, future research should focus on deploying sensor-based technologies for real-time monitoring of water quality (turbidity, E. coli, heavy metals) and flow dynamics, particularly in vulnerable areas like Bole, Akaki Kality, and Lake Beseka, to address data gaps and enhance IUWSI accuracy. Additionally, studies should quantify the economic impacts of water insecurity on households and agriculture, evaluate pilot interventions like nanotechnology and community-based governance, and develop transboundary frameworks.

Further research should explore climate adaptation strategies, such as rainwater harvesting, to counter projected warming and precipitation variability by 2030, and integrate Ethiopia's cultural frameworks, like kebele and iddir systems, to boost community engagement and address governance dissatisfaction.

Appendices

Appendix A: Questionnaire summary

- **Sample Size:** 385 (331 households, 38 community leaders, 16 AAWSA officials).
- **Sub-Cities:** Addis Ketema (38), Akaki Kality (61), Arada (13), Bole (50), Gullele (23), Kirkos (12), Kolfe Keranio (38), Lemi Kura (53), Lideta (13), Nifas Silk-Lafto (27), Yeka (27).
- **Downstream:** Adama (38), Awash Town (38), Lake Beseka (38).

Key metrics

73.3% piped water, 79.2% shortages, 62.4% treat water, 71.3% dissatisfied with AAWSA

Appendix B: AHP weighting example

- **Drinking Water Weights:** Availability (0.40), quality (0.30), piped water access (0.20), illnesses (0.10), derived from pairwise comparisons by 20 stakeholders.
- **Consistency Ratio:** <0.1, ensuring reliable weights.

Appendix C: ArcGIS mapping notes

- **Data Layers:** IUWSI scores, water availability, pollution (turbidity, E. coli, WQI), infrastructure issues.
- **Spatial Unit:** Sub-city and downstream boundaries, using AwashRiverBasin_Boundary.shp and Awash main rivers.shp.
- **Visualization:** Choropleth maps (red-to-green gradient) and kernel density for pollution hotspots.

Appendix D: Python code for analysis and visualization

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from scipy.stats import chi2_contingency
import statsmodels.api as sm
import matplotlib.pyplot as plt
import seaborn as sns
import geopandas as gpd
from matplotlib.backends.backend_pdf import PdfPages
import folium
from folium.plugins import HeatMap

# Data including 11 sub-cities and downstream areas
data = {
    'Location': ['Addis Ketema', 'Akaki Kality', 'Arada', 'Bole', 'Gullele', 'Kirkos', 'Kolfe Keranio',
    'Lemi Kura', 'Lideta', 'Nifas Silk-Lafto', 'Yeka', 'Adama', 'Awash Town', 'Lake Beseka'],
    'Piped_Water (%)': [90.0, 68.8, 100.0, 61.5, 66.7, 75.0, 80.0, 78.6, 100.0, 71.4, 85.7, 60.0, 50.0,
    40.0],
    'Availability_24_7 (%)': [20.0, 0.0, 0.0, 0.0, 0.0, 0.0, 10.0, 7.1, 0.0, 0.0, 0.0, 0.0, 0.0],
    'Frequent_Shortages (%)': [70.0, 87.5, 66.7, 92.3, 83.3, 75.0, 90.0, 85.7, 66.7, 85.7, 80.0, 80.0,
    90.0, 100.0],
    'Median_Water_Use (L)': [150, 75, 75, 75, 150, 50, 75, 150, 150, 150, 100, 60, 50, 40],
    'Median_Cost (ETB)': [250, 300, 200, 500, 250, 300, 300, 300, 250, 300, 275, 200, 200, 200],
    'IUWSI': [3.0, 2.2, 2.8, 2.1, 2.5, 2.4, 2.3, 2.5, 2.5, 2.3, 2.6, 2.3, 2.4, 1.8],
    'Drinking_Water': [3.2, 2.3, 3.0, 2.1, 2.6, 2.6, 2.4, 2.5, 2.7, 2.5, 2.7, 2.4, 2.5, 1.7],
    'Ecosystems': [2.8, 1.5, 2.7, 1.4, 2.5, 2.3, 2.1, 2.6, 2.4, 2.0, 2.4, 1.8, 1.7, 1.2],
    'Climate_Hazards': [3.0, 2.2, 2.8, 2.0, 2.4, 2.3, 2.3, 2.4, 2.5, 2.3, 2.5, 2.3, 2.4, 2.0],
    'Socio_Economic': [3.0, 2.3, 2.8, 2.3, 2.5, 2.2, 2.4, 2.5, 2.4, 2.5, 2.6, 2.4, 2.5, 2.2],
    'Contamination_Awareness (%)': [50.0, 68.8, 66.7, 76.9, 66.7, 50.0, 60.0, 64.3, 33.3, 57.1, 60.0,
    70.0, 80.0, 90.0],
    'Turbidity (NTU)': [5.0, 15.0, 5.0, 20.0, 10.0, 10.0, 15.0, 10.0, 5.0, 15.0, 10.0, 20.0, 25.0, 30.0],
    'E_coli (CFU/100 mL)': [10, 20, 10, 25, 15, 15, 20, 15, 10, 20, 15, 30, 40, 50],
    'WQI': [80, 60, 80, 55, 70, 70, 60, 70, 80, 60, 70, 50, 45, 40]
}
df = pd.DataFrame(data)
df.to_csv('arb_water_security.csv', index=False)

# Descriptive Statistics
print("Descriptive Statistics:\n", df.describe())
```

```
# Chi-Square Test: Location vs. Availability
contingency_table = pd.crosstab(df['Location'], df['Availability_24_7 (%)'])
chi2, p, _, _ = chi2_contingency(contingency_table)
print(f'Chi-Square Test (Location vs. Availability): chi2={chi2:.2f}, p={p:.3f}')

# Regression Analysis
X = df[['Piped_Water (%)', 'Frequent_Shortages (%)', 'Median_Cost (ETB)', 'Turbidity (NTU)',
'E_coli (CFU/100 mL)']]
y = df['IUWSI']
X = sm.add_constant(X)
model = sm.OLS(y, X).fit()
with open('regression_summary.txt', 'w') as f:
    f.write(model.summary().as_text())

# K-Means Clustering
features = df[['IUWSI', 'Drinking_Water', 'Ecosystems', 'Climate_Hazards', 'Socio_Economic']]
scaler = StandardScaler()
scaled_features = scaler.fit_transform(features)
kmeans = KMeans(n_clusters=3, random_state=42)
df['Cluster'] = kmeans.fit_predict(scaled_features)

# Create PDF for visualizations
with PdfPages('arb_water_quality_visualizations.pdf') as pdf:
    # Bar Chart: Piped Water Access
    plt.figure(figsize=(14, 6))
    sns.barplot(x='Location', y='Piped_Water (%)', data=df, palette='viridis')
    plt.xticks(rotation=45)
    plt.xlabel('Location')
    plt.ylabel('Piped Water Access (%)')
    plt.title('Piped Water Access Across the Awash River Basin')
    plt.tight_layout()
    pdf.savefig()
    plt.close()

# Box Plot: Water Quality Ratings
quality_data = {
    'Location': ['Addis_Ketema']*10 + ['Akaki_Kality']*16 + ['Arada']*3 + ['Bole']*13 +
    ['Gullele']*6 + ['Kirkos']*3 + ['Kolfe_Keranio']*10 + ['Lemi_Kura']*14 + ['Lideta']*3 + ['Nifas_Silk-
    Lafto']*7 + ['Yeka']*7 + ['Adama']*10 + ['Awash_Town']*10 + ['Lake_Beseka']*10,
```

```
'Quality_Rating': [3]*4 + [2]*3 + [4]*2 + [3]*1 + [1]*8 + [2]*6 + [3]*2 + [2]*4 + [1]*3 +
[2]*3 + [3]*3 + [2]*5 + [1]*5 + [3]*3 + [2]*2 + [3]*3 + [2]*2 + [2]*4 + [1]*4 + [2]*2 + [1]*6 +
[2]*2}
df_quality = pd.DataFrame(quality_data)
plt.figure(figsize=(14, 6))
sns.boxplot(x='Location', y='Quality_Rating', data=df_quality, palette='magma')
plt.xticks(rotation=45)
plt.xlabel('Location')
plt.ylabel('Water Quality Rating (1=Poor, 4=Excellent)')
plt.title('Water Quality Ratings Across the Awash River Basin')
plt.tight_layout()
pdf.savefig()
plt.close()

# Line Plot: IUWSI Projections
years = ['2025', '2030', '2035', '2040', '2050']
bau = [2.3, 2.0, 1.8, 1.7, 1.6]
sd = [2.3, 3.0, 3.4, 3.7, 3.9]
plt.figure(figsize=(10, 6))
plt.plot(years, bau, 'r-o', label='Business-as-Usual')
plt.plot(years, sd, 'g-^', label='Sustainable Development')
plt.xlabel('Year')
plt.ylabel('IUWSI Score')
plt.title('IUWSI Projections for the Awash River Basin (2025-2050)')
plt.legend()
plt.grid(True)
plt.tight_layout()
pdf.savefig()
plt.close()
# Water Quality Maps (IUWSI, Turbidity, E. coli, WQI)
gdf = gpd.GeoDataFrame(
    df,
    geometry=gpd.points_from_xy(
        df['Longitude'],[38.750, 38.730, 38.760, 38.790, 38.740, 38.770, 38.700, 38.780, 38.750,
38.720, 38.800, 39.270, 40.160, 40.750],
        df['Latitude'],[9.040, 8.955, 9.045, 9.015, 9.055, 9.020, 8.970, 9.000, 9.035, 8.990, 9.030,
8.540, 9.140, 10.500]
    )
)
for param, title in [
```

```

('IUWSI', 'IUWSI Across the Awash River Basin'),
('Turbidity (NTU)', 'Turbidity (NTU) Across the Awash River Basin'),
('E_coli (CFU/100 mL)', 'E. coli (CFU/100 mL) Across the Awash River Basin'),
('WQI', 'Water Quality Index Across the Awash River Basin')
]:
fig, ax = plt.subplots(figsize=(12, 8))
basin_boundary = gpd.read_file('AwashRiverBasin_Boundary.shp')
rivers = gpd.read_file('Awash main rivers.shp')
basin_boundary.plot(ax=ax, color='lightgrey', edgecolor='black')
rivers.plot(ax=ax, color='blue', linewidth=1)
gdf.plot(column=param, ax=ax, cmap='RdYlGn', markersize=50, legend=True,
         legend_kwds={'label': param, 'orientation': 'horizontal'})
for idx, row in gdf.iterrows():
    ax.annotate(row['Location'], xy=(row.geometry.x, row.geometry.y), fontsize=8)
plt.title(title)
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.tight_layout()
pdf.savefig()
plt.close()

# Save Folium map for interactive use
m = folium.Map(location=[9.5, 39.5], zoom_start=8)
HeatMap(data=[[row.geometry.y, row.geometry.x, row.IUWSI] for _, row in

```

Appendix E: Sampling stations

Station	Longitude	Latitude	IUWSI	Arsenic ($\text{\AA}\mu\text{g/L}$)	Manganese ($\text{\AA}\mu\text{g/L}$)	EOCs ($\text{\AA}\mu\text{g/L}$)	BOD (mg/L)
SW1	38.745	9.035	3	2.5	25	0.15	6
SW2	38.735	8.95	2.3	7.5	75	0.75	24
SW3	38.755	9.04	2.8	3.5	35	0.25	12
SW4	38.785	9.01	2	9.5	95	0.95	28
SW5	38.735	9.05	2.6	5.5	55	0.55	16
SW6	38.765	9.015	2.4	4.5	45	0.45	14
SW7	38.695	8.965	2.2	6.5	65	0.65	22
SW8	38.775	8.995	2.5	6	60	0.6	20
SW9	38.745	9.03	2.7	3	35	0.35	10
SW10	38.715	8.985	2.3	8.5	85	0.85	24
SW11	38.795	9.025	2.6	5	55	0.5	16
SW12	38.74	9.045	2.9	2	20	0.2	8
SW13	38.725	8.96	2.1	8	80	0.8	26
SW14	38.77	9.02	2.4	4	40	0.4	14
SW15	38.75	9.04	2.5	7	70	0.7	20
SW16	38.72	8.975	2.2	9	90	0.9	22
SW17	38.78	9.005	2.8	3.5	30	0.3	10
SW18	38.71	8.955	2	10	100	1	30
SW19	38.76	9.035	2.7	2.5	25	0.25	8
SW20	38.73	8.99	2.5	6	60	0.6	18

Appendix F: Water security

Location	Piped Water (%)	Availability 24_7 (%)	Frequency Short-ages (%)	Median Water Use (L)	Median Cost (ETB)	IUWSI	Drinking Water	Ecosystems	Climate Hazards	Socio-Economic	Contamination Awareness (%)	Turbidity (NTU)	E.coli (CFU /100 mL)	WQI	Longitude	Latitude
Addis Ketema	90	20	70	150	250	3	3.2	2.8	3	3	50	5	10	80	38.75	9.04
Akaki Kaliti	68.8	0	87.5	75	300	2.2	2.3	1.5	2.2	2.3	68.8	15	20	60	38.73	8.955
Arada	100	0	66.7	75	200	2.8	3	2.7	2.8	2.8	66.7	5	10	80	38.76	9.045
Bole	61.5	0	92.3	75	500	2.1	2.1	1.4	2	2.3	76.9	20	25	55	38.79	9.015
Gullele	66.7	0	83.3	150	250	2.5	2.6	2.5	2.4	2.5	66.7	10	15	70	38.74	9.055
Kirkos	75	0	75	50	300	2.4	2.6	2.3	2.3	2.2	50	10	15	70	38.77	9.02
Kolfe Keranio	80	10	90	75	300	2.3	2.4	2.1	2.3	2.4	60	15	20	60	38.7	8.97
Lemi Kura	78.6	7.1	85.7	150	300	2.5	2.5	2.6	2.4	2.5	64.3	10	15	70	38.78	9
Lideta	100	0	66.7	150	250	2.5	2.7	2.4	2.5	2.4	33.3	5	10	80	38.75	9.035
Nifas Silk-Lafto	71.4	0	85.7	150	300	2.3	2.5	2	2.3	2.5	57.1	15	20	60	38.72	8.99
Yeka	85.7	0	80	100	275	2.6	2.7	2.4	2.5	2.6	60	10	15	70	38.8	9.03

Abbreviations

AAWSA	Addis Ababa Water and Sewerage Authority
AHP	Analytic Hierarchy Process
ARB	Awash River Basin
a.s.l.	Above sea level
BAU	Business-as-Usual
BDA	Basin Development Authority
BOD	Biochemical Oxygen Demand
CDF	Cumulative Distribution Function
CFU	Colony Forming Units (as in CFU/100 mL for E. coli)
CMIP6	Coupled Model Intercomparison Project Phase 6
CS	Central Statistical Agency
DECS	Drinking Water and Human well-being, Ecosystems, Climate Change and Water-Related Hazards, Socio-Economic Aspects
E. coli	Escherichia coli (bacterial indicator)
EOCs	Emerging Organic Contaminants
ESS-WQT	Ethiopian Standards for Water Quality Test
ETB	Ethiopian Birr
GCM	General Circulation Model
GDP	Gross Domestic Product
HDPE	High-Density Polyethylene
IPCC	Intergovernmental Panel on Climate Change
ITCZ	Inter-Tropical Convergence Zone
IUWSI	Integrated Urban Water Security Index
MoWE	Ministry of Water and Energy
NTU	Nephelometric Turbidity Units (unit for turbidity measurement)
OLS	Ordinary Least Squares (regression method)
QM	Quantile Mapping
SD	Sustainable Development
SDG	Sustainable Development Goal
SSP	Shared Socioeconomic Pathway
Tmax	Maximum temperature
Tmin	Minimum temperature
UN-Water	United Nations Water
UWSI	Urban Water Security Index
WHO	World Health Organization
WQI	Water Quality Index

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Author contributions

Conceptualization; A.M, T.A., and S.G.; methodology; A.M; validation, T.A., S.G., L.T, and D.H.; formal analysis, A.M.; investigation, A.M.; resources, A.M; data curation, A.M. and G.B; writing original draft preparation, A.M., L.T.; writing review

and editing; T.A. D.H., L.T. and G.B. visualization, A.M, T.A, D.H. and S.G.; supervision, T.A. S.G and D.H.; project administration, T.A., S.G., L.T., and D.H; Software; A.M and G.B

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Data availability

The datasets used and/or analyzed during the current study are available on reasonable request to the corresponding author.

Declarations

Ethics approval and Consent to participate

The interview procedure of this study was approved by the Ethiopian Institute of Water Resources, Addis Ababa University and followed the relevant guidelines. Informed consent for participation was obtained from all participants. Informed consent was obtained from all participants involved in the study prior to data collection. For participants under the age of 18, informed consent was obtained from a parent and/or legal guardian.

Clinical trial number

Not applicable.

Conflicts of Interest

The authors declare no conflict of interest.

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